

bldc_controller_routed

Design Report

kicad-tools 0.13.0

Rev 1 | 2026-06-10 | jlcpb

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	55 (0 SMD, 0 THT, 55 other)
Nets	52
Traces	257 segments
Vias	51
Board Size	80.0 x 100.0 mm

Design Overview

Theory of Operation

BLDC Motor Controller

3-Phase Brushless DC Motor Driver

Thermal analysis and high-current routing demo

Power Architecture

Power Rails: +24V, +3V3, +5V, GND, PWR_FLAG

Regulator	Device
U1	LM2596-5.0
U2	AMS1117-3.3

Assembly Notes

1 fine-pitch component; 4 polarized components

- **Fine-pitch components:** 1 (U10)
- **Polarized components:** 4 – check orientation markings

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: SKIPPED – ERC skipped by user request

Bill of Materials

Value	Package	Qty	References
100nF	C_0805_2012Metric	7	C2, C7, C8, C12, C13, C14
10nF		3	C30, C31, C32
10uF	C_0805_2012Metric	3	C5, C6, C16
20pF		2	C10, C11
220uF	C_0805_2012Metric	2	C3, C4
4.7uF	C_0805_2012Metric	1	C9
470uF	C_0805_2012Metric	1	C1
PWR		1	D3
SMBJ24A	D_SMA	1	D1
SS34	D_SMA	1	D2
STATUS		1	D4
15A	Fuse_1206_3216Metric	1	F1
Hall Sensors	PinHeader_1x05_P2.54mm_Vertical	1	J3
Motor Output	PinHeader_1x03_P2.54mm_Vertical	1	J2
Power Input	PinHeader_1x02_P2.54mm_Vertical	1	J1
SWD-6		1	J4
33uH	L_1210_3225Metric	1	L1
IRLZ44N		6	Q1, Q2, Q3, Q4, Q5, Q6
10k		3	R30, R31, R32
1k		2	R3, R4
22		3	R20, R21, R22
5mR		3	R10, R11, R12
AMS1117-3.3	SOT-223-3_TabPin2	1	U2
DRV8301	HTSSOP-56-1EP_6.1x14mm_P0.5mm_EP3.61x6.35mm	1	U3
LM2596-5.0	TO-263-5_TabPin3	1	U1
STM32G431K8Tx	LQFP-32_7x7mm_P0.8mm	1	U10
8MHz		1	Y1

DRC Status

Metric	Count
Errors	9
Warnings	57
Blocking	9

Status: FAIL ### Violations by Type

Violation Type	Count
pad_grid	56
connectivity	22
single_pad_net	8
clearance_pad_segment	6
clearance_segment_via	2
clearance_segment_segment	1
zone_unfilled	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 9 blocking DRC violations (clearance_pad_segment (6), clearance_segment_via (2), clearance_segment_segment (1))
- **[OPTIONAL]** Verify zone fill in KiCad: 21 nets appear incomplete but may be connected via zone fills
- **[OPTIONAL]** Verify zone fill in KiCad for 6 zone-connected nets
- **[OPTIONAL]** Review 57 DRC warnings
- **[OPTIONAL]** Analog net: ISENSE_A+ — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: ISENSE_A- — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: ISENSE_B+ — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: ISENSE_B- — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: ISENSE_C+ — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: ISENSE_C- — analog signal; noise-sensitive, avoid crossing digital signals

Routing Status

Metric	Value
Signal Net Completion	44.7% (17/38)
Overall Completion	57.7%
Complete Nets	30 / 52
Zone-Connected Nets	6
Single-Pad Nets	8 (no routing needed)
Incomplete Nets	22
Unconnected Pads	68

Zone-Connected Nets

- +24V
- +3V3
- +5V
- GND
- PWR_LED
- VIN

Single-Pad Nets

8 single-pad nets (no routing needed) – not listed individually.

Unrouted Signal Nets

- BST_A
- BST_B
- BST_C
- GATE_AL
- GATE_BL
- GATE_CL
- HALL_A
- HALL_B
- HALL_C
- ISENSE_A+
- ISENSE_A-
- ISENSE_B-
- ISENSE_C-
- NRST
- OSC_IN
- PHASE_A
- PHASE_B
- PHASE_C
- PWM_AH
- PWM_BH
- PWM_CL

Unrouted Signal Nets

- BST_A
- BST_B
- BST_C
- GATE_AL
- GATE_BL
- GATE_CL
- HALL_A
- HALL_B
- HALL_C
- ISENSE_A+
- ISENSE_A-
- ISENSE_B-
- ISENSE_C-
- NRST
- OSC_IN
- PHASE_A
- PHASE_B
- PHASE_C
- PWM_AH
- PWM_BH
- PWM_CL

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~2.0 USD
Components (estimated)	~3.2 USD
Assembly (estimated)	~2.35 USD
Total (estimated)	~7.56 USD
Batch Quantity	5
Batch Total (estimated)	~37.78 USD